=> Machine Learning Assignment 0 <=
100 MARKS

Deadline

If you have any problems with this practical assignment, speak up well before the deadline!
Submit on RuConnected.

Task 0: My Linux [20 Marks]

Run the following commands in your Linux terminal, then neatly paste them into `a0.ipynb`:

a) `sudo dmidecode -t system`
b) `sudo dmidecode -t processor`
c) `sudo dmidecode -t memory`
d) `nvidia-smi`

Deliverables:
- `a0.ipynb` (saved output)

Task 1: Problem Framing [30 Marks]

For each of the following scenarios, state whether it is a **Classification** or a **Regression** problem and briefly justify your answer in a sentence.

a) Predicting the exact number of points a basketball player will score in their next game.
b) Determining whether a credit card transaction is fraudulent or legitimate.
c) Estimating the sale price of a house based on features such as square metres and number of bedrooms.
d) Categorising news articles into topics like Sports, Politics, or Technology.
e) Forecasting the temperature in Celsius for tomorrow.
f) A city wants you to predict, for each household, how many days next month the household will have no piped water (an integer count that is often 0, sometimes a few, rarely many). Decide whether to frame this as classification or regression. Do some research and justify in 1–2 sentences, including a final concrete reason that your choice could mislead the city's planners.

Deliverables:

- `a0.ipynb` (saved output)

Task 2: Metrics [20 Marks]

Consider the following results from a binary classification model designed to detect spam emails (where 'spam' is the positive class):

- **True Positives (TP):** 85
- **False Positives (FP):** 10
- **True Negatives (TN):** 150
- **False Negatives (FN):** 5

Using Python code, manually calculate the following performance metrics (without any libraries). You must show the formula used and the steps of your calculation for each metric.

a) Accuracy
b) Precision
c) Recall (Sensitivity)
d) F1-Score

Deliverables:

- `a0.ipynb` (saved output)

Task 3: My first real classifier [30 Marks]

Your task is to create a Jupyter Notebook that accomplishes the following steps:

a) **Load and Split Data**: Load the Iris dataset from Scikit-learn and split it into a training set (80%) and a testing set (20%). Use a random_state for reproducibility.
b) **Train Model**: Initialise and train a KNeighborsClassifier from Scikit-learn on the training data. Use the default value for *k*.
c) **Predict**: Use the trained model to make predictions on the test set.
d) **Evaluate**:
   - Calculate and print the *accuracy_score*.
   - Generate and display the *confusion_matrix* for your model's predictions.
e) **Analyse**: In a markdown cell, briefly comment on the performance of your classifier based on the accuracy and the confusion matrix.

Deliverables:

- `a0.ipynb` (saved output)